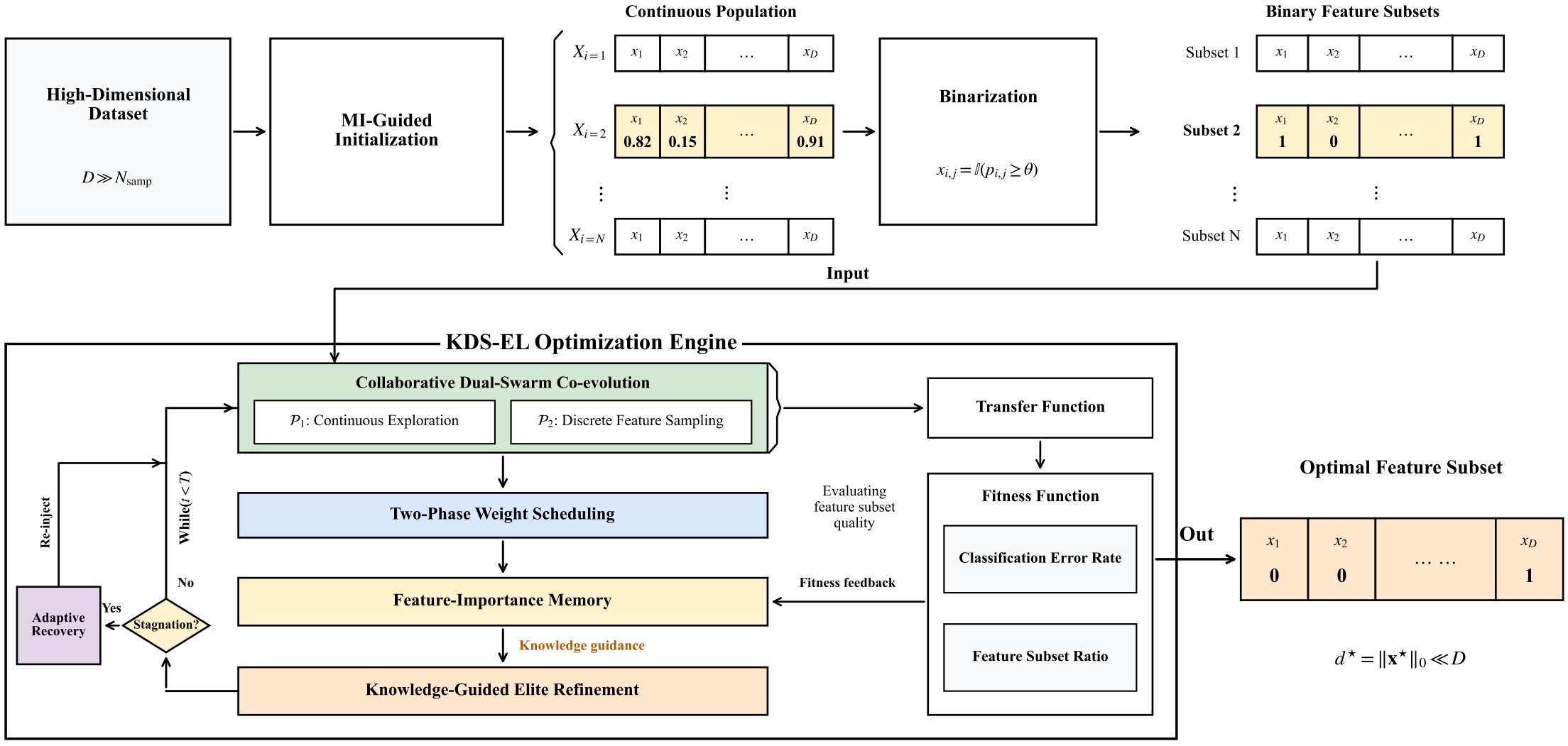

High-Dimensional Dataset
$D \gg N_{\text{samp}}$
MI-Guided Initialization
Continuous Population
$X_{i=1}$
$X_{i=2}$
$X_{i=N}$
$x_1$ $x_2$ … $x_D$
0.82 0.15 … 0.91
Binarization
$x_{i,j} = \mathbb{I}(p_{i,j} \geq \theta)$
Binary Feature Subsets
Subset 1
Subset 2
Subset N
1 0 … 1
Input
KDS-EL Optimization Engine
Collaborative Dual-Swarm Co-evolution
$\mathcal{P}_1$: Continuous Exploration
$\mathcal{P}_2$: Discrete Feature Sampling
Two-Phase Weight Scheduling
Feature-Importance Memory
Knowledge guidance
Knowledge-Guided Elite Refinement
Transfer Function
Fitness Function
Classification Error Rate
Feature Subset Ratio
Evaluating feature subset quality
Fitness feedback
Re-inject
While($t < T$)
No
Yes
Stagnation?
Adaptive Recovery
Out
Optimal Feature Subset
0 0 … … 1
$d^\star = \|\mathbf{x}^\star\|_0 \ll D$